



explain Simulation Review in details with substeps

Simulation Review is the checkpoint where you validate the design using virtual models (FEA/CFD/MBD/system simulations) before or in parallel with prototypes, to reduce risk and guide design changes. Below is a detailed, engineering-oriented breakdown with substeps.^[1]
^[2]

1. Define objectives and success criteria

- Clarify why you are running simulations: verify requirement compliance, compare design variants, size safety factors, or optimize performance and cost.^[2] ^[1]
- For each objective, define quantitative success criteria (e.g., max stress, deflection, pressure drop, temperature, factor of safety, stability margin) that map back to requirements.^[3] ^[2]

2. Plan the simulation strategy

- Decide which types of analysis are needed: structural (static, fatigue), thermal, CFD, vibration/modal, crash, kinematics, multi-body dynamics, system-level, etc.^[1] ^[2]
- For each type, define load cases, operating conditions, worst-case scenarios, and any transient events that the simulations must cover.^[4] ^[2]

3. Build or validate models

- Create or clean up the geometry, choose appropriate idealizations (symmetry, 2D vs 3D, shell vs solid, rigid vs flexible), and generate a suitable mesh.^[5] ^[2]
- Select material models and properties (nonlinearities, temperature dependence, damping, contact definitions) that reflect real behavior closely enough for the decisions you need.^[2] ^[3]

4. Set up loads, boundary conditions, and steps

- Apply realistic loads, constraints, and interactions (forces, pressures, flows, torques, contacts, thermal loads, motion inputs) based on use cases and test plans.^[5] ^[2]
- For nonlinear or time-dependent problems, define analysis “steps” and load “substeps” so loads ramp in a stable way and results capture critical events.^[6] ^[5]

5. Run simulations and ensure numerical quality

- Execute the analyses with appropriate solver settings (time step, substeps, convergence tolerances, contact settings), adjusting as needed for stability and performance.^{[2] [5]}
- Check numerical indicators: convergence behavior, energy balance, contact penetration, and sensitivity to mesh density or time-step size; refine where necessary.^{[6] [5]}

6. Post-process and interpret results

- Extract key outputs: stress/strain, displacements, factors of safety, flow patterns, pressure drops, temperatures, frequencies, modes, and response over time.^{[1] [2]}
- Compare results against the objectives and requirements, identifying hotspots, margins, unexpected behaviors, and possible failure modes.^{[4] [1]}

7. Correlate with physical data (when available)

- If prototype or historical test data exist, compare simulation predictions with measurements to judge model accuracy and adjust models or assumptions if needed.^{[7] [1]}
- Document correlation quality (good match, acceptable with caveats, poor) and what that implies for trust in predictions used for design decisions.^{[8] [7]}

8. Review assumptions, limitations, and risks

- List key modeling assumptions (simplified geometry, boundary conditions, ignored effects, idealized materials) and explain how they may bias results.^{[7] [1]}
- Identify remaining risks where simulation confidence is low or coverage is incomplete, and propose mitigation (extra tests, refined models, safety factors).^{[7] [4]}

9. Conduct the Simulation Review meeting

- Share a structured report beforehand: objectives, models, assumptions, load cases, key plots, tables of results vs requirements, and conclusions/recommendations.^{[1] [2]}
- In the review, walk stakeholders through each major simulation, highlight critical findings, and collect questions, challenges, and required follow-up actions.^{[7] [1]}

10. Decide design actions and maturity

- Based on the evidence, agree whether the design is acceptable, needs local reinforcement/optimization, or requires deeper architectural changes.^{[2] [1]}
- Prioritize actions (e.g., design modifications, additional simulations, physical tests) and define clear owners and deadlines before the next project milestone.^{[4] [7]}

11. Document and archive simulation data

- Store input files, result files, post-processing scripts, and review minutes in a controlled location with proper versioning and traceability to requirements.^[9] ^[7]
- Summarize key learnings and validated models so they can be reused in future projects and help standardize your simulation approach.^[10] ^[9]

If you want, I can now turn these Simulation Review substeps into a compact checklist (yes/no or scoring items) tailored for typical mechanical CAE workflows like FEA and CFD.



1. <https://www.twi-global.com/technical-knowledge/faqs/faq-what-is-simulation>
2. <https://www.synopsys.com/glossary/what-is-simulation.html>
3. <https://www.igi-global.com/dictionary/future-opportunities-for-using-gamification-in-management-education/26957>
4. <https://www.theknowledgeacademy.com/blog/what-is-simulation/>
5. https://ansyshelp.ansys.com/public/Views/Secured/corp/v252/en/wb_sim/ds_Steps_Substeps_Eq.html
6. https://ansyshelp.ansys.com/public/Views/Secured/corp/v251/en/ans_nlad/Hlp_G_ADVRZSUB.html
7. <https://www.sciencedirect.com/science/article/pii/S0951832015000307>
8. https://ne.ncsu.edu/wp-content/uploads/2017/07/04_Kaizer-VV-2017.pdf
9. [https://inacsl.memberclicks.net/assets/Review Tool for Best Simulation Practices \(JBP\) .pdf](https://inacsl.memberclicks.net/assets/Review%20Tool%20for%20Best%20Simulation%20Practices%20(JBP).pdf)
10. [https://eprints.whiterose.ac.uk/134849/1/An overview of realist evaluation for simulation-based education.pdf](https://eprints.whiterose.ac.uk/134849/1/An%20overview%20of%20realist%20evaluation%20for%20simulation-based%20education.pdf)
11. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10599702/>
12. <https://library.nshealth.ca/Simulation-Learning/Evaluation>
13. https://www.reddit.com/r/Houdini/comments/1kw69qz/effective_simulation_technique/
14. <https://github.com/rlguy/Blender-FLIP-Fluids/wiki/Domain-Stats-Settings>
15. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8861914/>